

Colorectal Polyps

Colorectal cancer and polyps (adenomas): high-effect genes (APC, MUTYH, CHEK2), common loci (8q24, SMAD7, BMP4, GREM1), polygenic scores and what this implies for the age and interval of colonoscopy

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	8
6. Favourable	9
7. Unfavourable	10
8. Mechanisms of ageing	11
9. Practical suggestions	12
10. Limitations	13
11. Methods and sources	14
12. Annex: all variants analysed	15

1. Summary

Favourable. There are no variants of high effect: APC I1307K, MUTYH Y179C and G396D and CHEK2 I157T are absent, so there is no sign of polyposis MUTYH or monogenic risk detected by the chip. It has 1 allele **MTHFR 677T** (lightly protective with adequate folate) and does not have the risk alleles of **EIF3H** or **GREM1**. The two small scores already used in §24 remain typical (55.6 and 48.7 percentiles).

Unfavourable. The most complete scores are above the median: that of 95 variants (Huyghe 2018) is in percentile 94.3 and the genomic of 1 million variants (Thomas 2023) in the 78.5. Early-onset colorectal cancer is in 82.3, colon polyps in 73.9 and benign colon neoplasia (D12) in 74.7. The loci load is high for both colorectal cancer (83.1) and polyps and adenomas (92.7). There are risk alleles in **BMP4** (2 copies), **8q24/MYC**, **SMAD7**, **COLCA2**, **10p14** and **BMP2** (1 copies each).

What does it mean? Without high effect genes, the risk is polygenic and moderately increased: the most complete scores put Sérgio F approximately in the upper quintile, with polyps above average. This updates the reading of §24, which only used small (typical) scores. The practical implication is not to postpone the screening: first quality colonoscopy at 45 years later, or already, if this age has passed. If there is a family history, earlier. The next interval should be decided by the findings (see Suggestions). Family history continues to weigh more than genes.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality DR2 ≥ 0.8 . Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 13 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 13 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 2.506.302 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population (n = 633) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVABLE/ALTO; ≤ 20 : MENOS PROVABLE/BAIXO; between 20 and 80: TIPIC.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 510^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population (n = 633) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 637 loci, 576 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk. The first two are the §24 (Canker Risk), maintained with the same values. The most complete scores (95 variants and 1 million variants) better capture the polygenic risk.

Colorectal cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Colorectal cancer	PGS000055 (Schmit 2019, 76 variants)	74 76 (weighted coverage 97%)	55,6	TIPIC	typical
Colorectal cancer	PGS000074 (Graff 2021, 103 variants)	100 103 (weighted coverage 93%)	48,7	TIPIC	typical
Colorectal cancer	PGS000765 (Huyghe 2018, 95 variants)	92 95 (weighted coverage 90%)	94,3	MAIS PROVABLE	unfavourable
Colorectal cancer	PGS003852 (Thomas 2023, LDpred2, 1.02 million variants)	988.504 1.016.596 (weighted coverage 97%)	78,5	TIPIC	typical
Early-onset colorectal cancer (< 50 years)	PGS003760 (Wang 2023, 49 variants)	48 49 (weighted coverage 98%)	82,3	MAIS PROVABLE	unfavourable

- **Colorectal cancer:** In §24 (Canker Risk).
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Polyps and adenomas

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Colonic polyps	PGS005151 (Jung 2025, CT-SLEB, 496 thousand variants)	481.405 496.139 (weighted coverage 96%)	73,9	TIPIC	typical
Benign neoplasia of the colon, rectum and anus (D12)	PGS004445 (Jung 2024, LDpred2, 1.06 million variants)	1.036.079 1.059.939 (weighted coverage 97%)	74,7	TIPIC	typical

4. Key variants by topic

High-effect variants (syndromes) and common replicated loci of colorectal cancer and adenomas. The genotypes are the same as §24.

High and moderate effect genes

Rare or moderate effect variants that greatly change the risk (polyposis MUTYH, APC I1307K, CHEK2). If present, they justify colonoscopy earlier and more frequent.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
APC	rs1801155	I1307K	T/T	chip	A (0 copy/s)	favourable	I1307K: creates an unstable treatment of adenins in the APC; colorectal risk ~1.5–2x	Strong
MUTYH	rs34612342	Y179C	T/T	chip	C (0 copy/s)	favourable	Y179C: loss of 8-oxo-G repair glycosylase; in homozygous or G396D causes polyposis MUTYH; heterozygous mild risk	Strong
MUTYH	rs36053993	G396D	C/C	chip	T (0 copy/s)	favourable	G396D: second variant of MUTYH polyposis	Strong
CHEK2	rs17879961	I157T	A/A	chip	G (0 copy/s)	favourable	I157T: CHEK2 reduced function; colorectal risk ~1.5x	Strong

Common loci of colorectal cancer and adenomas

Common loci of GWAS in the Wnt/MYC and TGF- β /BMP pathways, which control the renewal of colon crypts. Each one changes little the risk (~1.1–1.25x per allele), but add up.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MYC/CCAT2 (8q24)	rs6983267		G/T	chip	G (1 copy/s)	unfavourable (light)	The G allele increases the binding of TCF4 to MYC enhancer (via Wnt); colorectal risk ~1.2x per copy	Strong
EIF3H (8q23)	rs16892766		A/A	chip	C (0 copy/s)	favourable	The C allele is associated with colorectal risk ~1.25x per copy	Strong
SMAD7	rs4939827		T/C	chip	T (1 copy/s)	unfavourable (light)	The T allele lowers SMAD7 (via TGF- β /BMP); colorectal risk ~1.15x per copy	Strong
COLCA2 (11q23)	rs3802842		C/A	chip	C (1 copy/s)	unfavourable (light)	The C allele is associated with colorectal risk ~1.1x per copy	Moderate
GREM1/SCG5	rs4779584		C/C	chip	T (0 copy/s)	favourable	The T allele increases GREM1 (BMP antagonist); colorectal risk ~1.2x and adenomas	Moderate
10p14	rs10795668		G/A	chip	G (1 copy/s)	unfavourable (light)	The G allele is associated with colorectal risk ~1.1x (the A is protective)	Moderate
BMP4	rs4444235		C/C	chip	C (2 copy/s)	unfavourable	The C allele is associated with colorectal risk ~1.1x per copy	Moderate
BMP2 (20p12)	rs961253		C/A	chip	A (1 copy/s)	unfavourable (light)	Allele A is associated with colorectal risk ~1.1x per copy	Moderate

Folate and methylation

Folate and methylation of DNA influence the synthesis and repair of DNA in the colon epithelium.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MTHFR	rs1801133	C677T	G/A	chip	A (1 copy/s)	favourable	C677T (A in strand +): TT is associated with ~15-20% less colorectal cancer when folate intake is good	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) of colorectal cancer (colon, rectum, advanced cancer or adenoma) and colon polyps and adenomas. Alleles are oriented towards more risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Colorectal cancer	558 (507)	493 de 1014	504	83,1	unfavourable
Polyps and adenomas	79 (69)	76 de 138	69	92,7	unfavourable

6. Favourable

- **No variants of high effect:** APC I1307K T/T, MUTYH Y179C T/T and G396D C/C, CHEK2 I157T A/A. No sign of polyposis MUTYH. Strong for the tested variants.
- **EIF3H rs16892766 A/A** and **GREM1/SCG5 rs4779584 C/C**: without the risk alleles of two strong loci. **Moderate.**
- **MTHFR C677T G/A**: 1 allele T, slightly protective with good folate intake (the effect is mainly on T/T). **Weak.**
- **Small scores of typical §24**: percentiles 55.6 (76 variants) and 48.7 (103 variants). **Moderate.**

7. Unfavourable

- **Colorectal cancer, complete scores:** 95 variants in percentile 94.3 and 1 million variants in the 78.5; early onset in the 82.3. **Moderate.**
- **Colonic polyps** in percentile 73.9 and benign colon neoplasia in 74.7; load of polyps loci and adenomas in 92.7. **Moderate.**
- **Colorectal cancer loci load** in 83.1 (558 loci). **Moderate.**
- **BMP4 rs4444235 C/C** (2 alleles) and 1 allele in 8q24/MYC rs6983267, SMAD7 rs4939827, COLCA2 rs3802842, 10p14 rs10795668 and BMP2 rs961253. Moderate (each one ~ 1.1–1.2×).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Intercellular Communication	Wnt/MYC and TGF-β/BMP paths of crypts (BMP4 C/C, SMAD7, 8q24, BMP2)	Unfavourable (mild)
Somatic Mutation Accumulation	Adenomas arise from somatic mutations in APC and KRAS; polygenic risk of polyps in percentile 73.9	Unfavourable
DNA Damage and DNA Repair Deficit	MUTYH (8-oxo-guanina repair) and CHEK2 normal	Favourable
Genomic instability	No variants of high effect; the risk is polygenic	Neutral
Dysbiosis	Microbiota (fiber, butyrate, genotoxic bacteria) modulates the risk of polyps; see §42	Conditional
Chronic Inflammation	Mucosal inflammation accelerates the passage of adenoma to carcinoma	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **First quality colonoscopy at 45 years later** (or already, if this age has passed and has not yet been done), in a center with good rate of detection of adenomas and complete intestinal preparation. If there is a first-degree family member with colorectal cancer or advanced adenoma, start at 40 years or 10 years before the diagnosis of this family member (what comes first). *Suggestion (not applied)*.
2. **Interval after the first colonoscopy** (current orientations, adjusted to moderate polygenic risk):
3. normal colonoscopy: repeat at 10 years, or at 5-7 years by polyps and colorectal cancer scores above the average;
4. 1-2 adenomas < 10 mm: 7-10 years;
5. 3-4 adenomas < 10 mm: 3-5 years;
6. 5 or more adenomas, adenoma $\geq$ 10 mm, or adenoma with high grade dysplasia or villous component: 3 years;
7. Serrated polyps $\geq$ 10 mm: 3 years.

Suggestion (not applied). 3. Among colonoscopies: immunochemical test of occult blood in feces (FIT) annual or 2 in 2 years, and colonoscopy as soon as it is positive. Alert signs that justify colonoscopy at any age: blood in the feces, persistent change in intestinal transit, anemia due to lack of iron or fatigue without cause. Suggestion (not applied). 4. Fiber: the better ($\geq$ 50 g/day; legumes, whole grains, psyllium, PHGG) and daily crucifers. The fiber feeds butyrate-producing bacteria, which protect the epithelium of the colon (§42). Suggestion (not applied). 5. Red meat $\leq$ 300 g/week and zero processed meats; avoid much burned meat (heterocyclic amines, with the slow NAT2 of §24). Zero alcohol (already fulfills). Suggestion (not applied). 6. In the plane and protectors: vitamin D 10 000 IU (maintaining optimal levels), calcium from food, aerobic and strength exercise, and ferritin below 100 (without supplementary iron). Suggestion (not applied). 7. Low dose aspirin: reduces adenomas and colorectal cancer, but increases bleeding, especially with omega-3 3.2-4g/day. It only makes sense to discuss with the doctor if adenomas arise. Suggestion (not applied). 8. Informative: the class GLP-1 (retathrutide) is associated with less colorectal cancer in observational studies, and mTOR inhibition (rapamycin) reduces polyps in animal models. None of these substitutes screening. Suggestion (not applied). 9. Family history: asking and registering cases of colorectal cancer, endometrial cancer and polyps in the family. With Lynch syndrome or polyposis criteria, make a clinical germinal panel, because the chip does not read the genes MLH1, MSH2, MSH6 and PMS2 completely. *Suggestion (not applied)*.

10. Limitations

- The chip tests few variants of high effect. Lynch syndrome (MLH1, MSH2, MSH6, PMS2, EPCAM), familial adenomatous polyposis (APC) and other rare syndromes cannot be excluded; the clinical germinal panel and family history are essential.
- The scores disagree: the small of §24 are typical and the complete above the median. The complete ones have, in general, better performance, and the loci point in the same direction.
- The polyps scores (Jung 2024 and 2025) are based on health records and have little validated performance.
- The ages and ranges suggested are based on the population guidelines; the screening models adjusted to the polygenic risk are not yet in the official standards.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palyndromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** GRCh38 files from PGS Catalog (7 scores: 5 of colorectal cancer, including the 2 of §24 and 1 of early onset, and 2 of benign polyps/neoplasia). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510-8$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** Huyghe 2018 and Fernandez-Rozadilla 2023 (GWAS), Thomas 2020 and 2023 (scores), Jeon 2018 (tracking adjusted to polygenic risk), Gupta 2020 (intervals after polypectomy), Hampel 2010 (Lynch), Win 2014 (MUTYH), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
High and moderate effect genes	APC	rs1801155	T/T	chip	A	0	strand + (palindromic; assumed +)	without the risk allele
High and moderate effect genes	MUTYH	rs34612342	T/T	chip	C	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
High and moderate effect genes	MUTYH	rs36053993	C/C	chip	T	0	strand +	without the risk allele
High and moderate effect genes	CHEK2	rs17879961	A/A	chip	G	0	strand +	without the risk allele
Common loci of colorectal cancer and adenomas	MYC/CCAT2 (8q24)	rs6983267	G/T	chip	G	1	strand +	1 copy of risk allele
Common loci of colorectal cancer and adenomas	EIF3H (8q23)	rs16892766	A/A	chip	C	0	strand +	without the risk allele
Common loci of colorectal cancer and adenomas	SMAD7	rs4939827	T/C	chip	T	1	strand +	1 copy of risk allele
Common loci of colorectal cancer and adenomas	COLCA2 (11q23)	rs3802842	C/A	chip	C	1	strand +	1 copy of risk allele
Common loci of colorectal cancer and adenomas	GREM1/SCG5	rs4779584	C/C	chip	T	0	strand +	without the risk allele
Common loci of colorectal cancer and adenomas	10p14	rs10795668	G/A	chip	G	1	strand +	1 copy of risk allele
Common loci of colorectal cancer and adenomas	BMP4	rs4444235	C/C	chip	C	2	strand +	2 risk allele copies
Common loci of colorectal cancer and adenomas	BMP2 (20p12)	rs961253	C/A	chip	A	1	strand +	1 copy of risk allele
Folate and methylation	MTHFR	rs1801133	G/A	chip	A	1	strand +	1 copy of the protective allele